PYTHON

CONTROL STATEMENTS PROBLEMS

1. Write a Python program to find the largest of three numbers using conditional statements (without using built-in functions like max()).

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))

if a >= b and a >= c:
    print("Largest:", a)
elif b >= a and b >= c:
    print("Largest:", b)
else:
    print("Largest:", c)
```

OUTPUT:

```
Enter first number: 43
Enter second number: 53
Enter third number: 65
Largest: 65
```

2. Write a program to check whether a given year is a leap year or not. (Hint: divisible by 4, not by 100 unless also divisible by 400)

```
year = int(input("Enter year: "))

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Leap Year")
else:
    print("Not a Leap Year")
```

OUTPUT:

```
Enter year: 2003
Not a Leap Year
```

3. Write a program to check whether a number is positive, negative or zero.

```
num = int(input("Enter number: "))

if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")
```

OUTPUT:

```
Enter number: 52
Positive
```

4. Write a program to check whether a given character is a vowel or consonant.

```
ch = input("Enter a character: ").lower()

if ch in 'aeiou':
    print("Vowel")
else:
    print("Consonant")
```

OUTPUT:

```
Enter a character: N
Consonant
```

5. Write a program that takes two numbers and an operator (+, -, *, /) and performs the corresponding operation using conditional statements.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
op = input("Enter operator (+, -, *, /): ")

if op == '+':
    print("Result:", a + b)
elif op == '-':
    print("Result:", a - b)
elif op == '*':
    print("Result:", a * b)
elif op == '/':
    print("Result:", a / b)
else:
    print("Invalid operator")
```

OUTPUT:

```
Enter first number: 42
Enter second number: 32
Enter operator (+, -, *, /): *
Result: 1344
```

6. Write a program to calculate electricity bill based on units consumed:

- First 100 units → ₹5/unit
- Next 100 units → ₹7/unit
- Above 200 units → ₹10/unit

```python
units = int(input("Enter units consumed: "))

if units <= 100:
    bill = units * 5
elif units <= 200:
    bill = 100 * 5 + (units - 100) * 7
else:
    bill = 100 * 5 + 100 * 7 + (units - 200) * 10

print("Electricity Bill:", bill)
```

OUTPUT:

```
Enter units consumed: 250
Electricity Bill: 1700
```

- If the units are within 100, then the entire bill is calculated by multiplying the units directly by 5.
- since the rate for the first 100 units is ₹5 per unit. If the units exceed 100 but are less than or equal to 200, then the first 100 units are calculated at ₹5 per unit, and this amount is fixed. After that, the remaining units (that is, total units minus 100) are calculated at ₹7 per unit, and both amounts are added to get the total bill.
- If the units are greater than 200, then the first 100 units are charged at ₹5 per unit, the next 100 units are charged at ₹7 per unit, and these two amounts are fixed portions. Then, the remaining units beyond 200 (that is, total units minus 200) are calculated at ₹10 per unit. Finally, all three amounts are added together to get the total electricity bill.

7. Given three angles, check whether they can form a valid triangle. (Hint: Sum of angles must be 180)

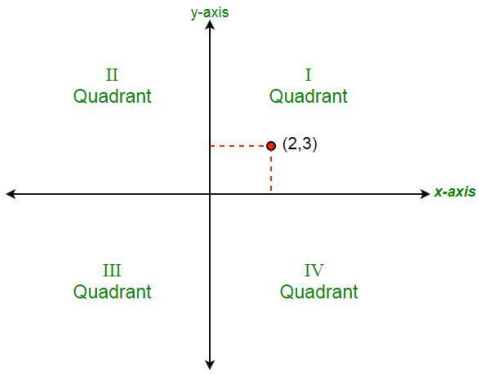
```python
a = int(input("Enter angle 1: "))
b = int(input("Enter angle 2: "))
c = int(input("Enter angle 3: "))

if a + b + c == 180:
    print("Valid Triangle")
else:
    print("Invalid Triangle")
```

OUTPUT:

```
Enter angle 1: 30
Enter angle 2: 60
Enter angle 3: 90
Valid Triangle
```

8. Write a program to determine the quadrant of a point (x, y) in a coordinate system.




```
x = int(input("Enter x: "))
y = int(input("Enter y: "))

if x > 0 and y > 0:
    print("Quadrant I")
elif x < 0 and y > 0:
    print("Quadrant II")
elif x < 0 and y < 0:
    print("Quadrant III")
elif x > 0 and y < 0:
    print("Quadrant IV")
else:
    print("On Axis")
```

OUTPUT:

```
Enter x: 4
Enter y: 9
Quadrant I
```

9. Write a program to assign grade based on marks and attendance:

- Marks ≥ 90 and attendance ≥ 75 → A
- Marks ≥ 75 → B
- Marks ≥ 50 → C
- Else → Fail

```
marks = int(input("Enter marks: "))
attendance = int(input("Enter attendance (%): "))

if marks >= 90 and attendance >= 75:
    print("Grade A")
elif marks >= 75:
    print("Grade B")
elif marks >= 50:
    print("Grade C")
else:
    print("Fail")
```

OUTPUT:

```
Enter marks: 87
Enter attendance (%): 78
Grade B
```

10. Check if a password is strong based on conditions:

- Minimum 8 characters
- Contains at least one digit
- Contains at least one uppercase letter

```
password = input("Enter password: ")

has_digit = any(char.isdigit() for char in password)
has_upper = any(char.isupper() for char in password)

if len(password) >= 8 and has_digit and has_upper:
    print("Strong Password")
else:
    print("Weak Password")
```

OUTPUT:

```
Enter password: #Rise26Sh
Strong Password
```

11. Write a program to simulate ATM withdrawal with conditions:

- Balance should be sufficient
- Withdrawal amount must be multiple of 100
- Minimum balance should be maintained (e.g., ₹1000)

```
balance = float(input("Enter balance: "))
amount = float(input("Enter withdrawal amount: "))

if amount % 100 != 0:
    print("Amount must be multiple of 100")
elif amount > balance:
    print("Insufficient balance")
elif balance - amount < 1000:
    print("Minimum balance must be maintained")
else:
    print("Transaction successful")
    print("Remaining balance:", balance - amount)
```

OUTPUT:

```
Enter balance: 6789
Enter withdrawal amount: 4000
Transaction successful
Remaining balance: 2789.0
```

12. Write a program to determine the winner between two players based on inputs: rock, paper, or scissors.

```
p1 = input("Player 1: ").lower()
p2 = input("Player 2: ").lower()

if p1 == p2:
    print("Draw")
elif (p1 == "rock" and p2 == "scissors") or \
     (p1 == "paper" and p2 == "rock") or \
     (p1 == "scissors" and p2 == "paper"):
    print("Player 1 wins")
else:
    print("Player 2 wins")
```

\ is used for new line → backslash

OUTPUT:

```
Player 1: rock
Player 2: paper
Player 2 wins
```

13. Write a program to calculate profit or loss:

- If selling price > cost price → Profit
- If selling price < cost price → Loss
- Else → No profit, no loss

```
cp = float(input("Enter cost price: "))
sp = float(input("Enter selling price: "))

if sp > cp:
    print("Profit:", sp - cp)
elif sp < cp:
    print("Loss:", cp - sp)
else:
    print("No Profit No Loss")
```

OUTPUT:

```
Enter cost price: 568.8
Enter selling price: 600.0
Profit: 31.200000000000045
```

14. Check whether a number is divisible by both 3 and 5, only 3, only 5, or neither.

```
num = int(input("Enter number: "))

if num % 3 == 0 and num % 5 == 0:
    print("Divisible by both 3 and 5")
elif num % 3 == 0:
    print("Divisible by 3 only")
elif num % 5 == 0:
    print("Divisible by 5 only")
else:
    print("Not divisible by both")
```

OUTPUT:

```
Enter number: 567
Divisible by 3 only
```

15. Check whether a character is alphabet, digit, or special character.

```
ch = input("Enter a character: ")

if ch.isalpha():
    print("Alphabet")
elif ch.isdigit():
    print("Digit")
else:
    print("Special Character")
```

OUTPUT:

```
Enter a character: 87
Digit

Enter a character: 6%
Special Character
```